

FONG CHI (KEN) HO

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PhD Candidate in Electrical and Computer Engineering

Photon-Counting CT • Imaging Phantoms • Quantitative Image Quality • AI/ML

RESEARCH PROFILE

Medical imaging researcher specializing in photon-counting CT, physical and computational phantoms, experimental imaging systems, task-based image-quality assessment, and AI-enabled quantitative imaging. Experience spans digital development of an anthropomorphic lung-disease phantom fabricated through 3D printing, quantitative validation of the resulting physical phantom, construction of a benchtop dual-energy photon-counting CT system, and virtual imaging trials for technology evaluation.

EDUCATION

Ph.D., Electrical and Computer Engineering, Duke University	Expected May 2027
M.S., Electrical Engineering, University of California, Irvine	2022
B.S., Electrical Engineering, University of California, Los Angeles	2019

RELEVANT RESEARCH EXPERIENCE

Center for Virtual Imaging Trials, Duke University	Durham, NC
Research Assistant	Aug 2022–Present

- Developed the digital model for an anthropomorphic lung-disease phantom, which was physically fabricated through 3D printing by collaborators, and quantitatively evaluated the resulting phantom using photon-counting CT through a Canon Medical and an FDA U01-supported projects.
- Developed patient-specific 3D full-generation airway models and virtual imaging trials for COPD, including chronic bronchitis, mucus plugging, airway wall remodeling, distortion, and pruning.
- Performed task-specific digital and physical phantom evaluation across dose, spatial-resolution mode, reconstruction algorithm, and denoising strength using image quality and quantitative CT biomarkers.
- Conducted energy-integrating and photon-counting CT simulations using DukeSim, ray tracing, Monte Carlo methods, and SLURM-based HPC workflows to assess image quality, quantification accuracy, and protocol harmonization.
- Contributed to the XCAT-3.0 digital-twin library, Virtual NLST, and AAPM Task Group 387 recommendations for reliable computational phantom representation and interoperability.

Imaging Physics Laboratory, University of California, Irvine	Irvine, CA
Research Assistant	Jun 2021–Aug 2022

- Built and integrated a benchtop dual-energy photon-counting breast cone-beam CT system for microcalcification imaging, material decomposition, and detector spectrum-distortion correction.
- Calibrated the X-ray tube, rotation stage, power supply, and scanner components; developed data-acquisition, image-reconstruction, analysis, and graphical-interface pipelines supporting experimental imaging from raw measurements to 3D volumes.
- Implemented postmortem breast material decomposition and deep-learning/per-pixel spectrum-fitting methods for CdTe photon-counting detector correction.

SELECTED RESEARCH PROJECTS

NIH R01 HL155293 — Accuracy and Precision in CT Quantification of COPD Through Virtual Imaging Trials

FDA U01 FD007987 via FLUIDDA subaward — Development of a human lung-disease phantom for 3D printing and quantitative imaging validation.

Canon Medical Research — Photon-counting CT acquisition and postprocessing, realistic phantom evaluation, and deep-learning reconstruction assessment.

GE HealthCare sponsored research — Evaluation of novel and deep-learning reconstruction algorithms and spectral imaging for lung and bone quantification.

TECHNICAL EXPERTISE

Imaging and phantoms: Photon-counting CT, energy-integrating CT, cone-beam CT, experimental system integration and calibration, digital anthropomorphic phantom development, evaluation of externally fabricated 3D-printed phantoms, computational phantoms, virtual imaging trials, image reconstruction, material decomposition, quantitative image-quality assessment.

Programming and engineering: Python, MATLAB, C++, Julia, CAD, ImageJ, DICOM, Git, Linux, HPC/SLURM, ray tracing, Monte Carlo simulation, deep learning, computer vision, statistical analysis.

SELECTED PUBLICATIONS AND MANUSCRIPTS

- **Ho FC, Kavuri A, Sotoudeh-Paima S, et al.** CT performance in quantifying small airway disease using paired inspiratory–expiratory scans: A virtual imaging study. **Under review.**
- **Ho FC, Mei K, Sharma S, et al.** Task-specific evaluation of photon-counting CT using a 3D-printed anthropomorphic lung phantom with COPD pathology. **In preparation.**
- **Ho FC, Zarei M, Bache S, et al.** Quantification accuracy, consistency, and image quality in deep-learning CT reconstruction kernels optimized for pulmonary imaging. **Under review.**
- **Ho FC, Segars WP, Samei E, Abadi E.** Subject-specific airway modeling of patients with chronic bronchitis and small airway disease. **Under review.**
- **Tushar FI, Vancoillie L, McCabe C, et al.** Virtual lung screening trial: An in silico study inspired by the National Lung Screening Trial for lung cancer detection. *Medical Image Analysis*. 2025;103:103576.
- **Dahal L, Ghoghnejad M, Vancoillie L, et al.** XCAT-3.0: A comprehensive library of personalized digital twins derived from CT scans. *Medical Image Analysis*. 2025;103:103636.

SELECTED CONFERENCE PROCEEDINGS AND PRESENTATIONS

- **Ho FC, Mei K, Sharma S, et al.** Task-specific evaluation of photon-counting CT using a 3D-printed anthropomorphic lung phantom with COPD pathology. *RSNA Scientific Assembly*, 2025.
- **Ho FC, Zarei M, Bache S, et al.** Quantification accuracy, consistency, and image quality in deep-learning CT reconstruction kernels optimized for pulmonary imaging. *RSNA Scientific Assembly*, 2025.
- **Ho FC, Kavuri A, Sotoudeh-Paima S, et al.** CT performance in quantifying small airway disease using paired inspiratory–expiratory scans. *Virtual Imaging Trials in Medicine*, 2024.
- **Ho FC, Segars WP, Samei E, Abadi E.** Subject-specific airway modeling of patients with chronic bronchitis and small airway disease. *Virtual Imaging Trials in Medicine*, 2024.
- **Ho FC, Sotoudeh-Paima S, Segars WP, et al.** Development and application of a virtual imaging trial framework for airway quantifications via CT. *SPIE Medical Imaging*, 2023.

ADDITIONAL PROFESSIONAL EXPERIENCE

Office of Translation & Commercialization, Duke University

Durham, NC

Technology and Intellectual Property Fellow

Aug 2025–Present

- Evaluate university inventions through prior-art, competitive-landscape, market, and commercialization analyses; prepare technical assessment reports and industry-facing technology summaries.

OhmMaker LLC

Remote

Associate Engineer in Machine Learning

May–Jun 2020

- Developed computer-vision and speech-recognition models using transfer learning, GANs, CNNs, RNNs, LSTMs, and GRUs; designed mixed-signal PCB schematics and layouts.

Topnet Inc.

Los Angeles, CA

Data Engineer and Sales Administrator

Nov 2019–May 2020

- Built data pipelines, dashboards, and a computer-vision inventory application using Python, SQL, Airflow, Spark, Hadoop, and Tableau.

TEACHING AND MENTORSHIP

Undergraduate Research Advisor, Duke Muser Program (2023–Present) — Mentored four undergraduate researchers in quantitative CT and generative AI projects.

Teaching Assistant, Duke ECE 110L (Spring 2024) — Supported Fundamentals of Electrical and Computer Engineering laboratory instruction.

REFEREES

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